

Response to Referee R6

We thank the editor and referees for their continued engagement with our paper. Below we provide a brief summary of the main changes in this revision and then respond to each comment in turn.

- **Stock-return analysis.** We thank R2 for suggesting a fuller treatment of the stock market evidence, including events beyond those the previous draft emphasized. That examination identified an additional event that was meaningful for gold clauses: the November 1934 window, in which the second certiorari grant coincided with the midterm election, now part of the main analysis. The evidence appears in a dedicated table (Table 1, in the main text) and four event-study figures (Internet Appendix Figures IA.2–IA.5). Table 1 reports, for every main event window (the Joint Resolution, the November 1934 certiorari grant and midterm election, the Supreme Court oral arguments and post-argument days, and the decision, plus the full May 1933–February 1935 episode), the cumulative returns of equal-weighted gold-exposed and non-exposed portfolios—pooled and within size terciles formed on end-1932 market capitalization—together with raw and beta-adjusted gold-minus-non-gold differentials and t -statistics. The figures plot the four underlying return series (the value-weighted market, the two equal-weighted portfolios, and the gold-exposure-weighted portfolio) around each event. Over the Joint Resolution window, gold-exposed firms returned 30.9% against 25.4% for non-exposed firms, a differential of +5.5 percentage points ($t \approx 3.0$). A new Internet Appendix section examines every intermediate judicial event of the litigation (the first hearing, the lower-court rulings, and the first certiorari grant) in the same format (Table IA.20); none produced a significant differential.
- **Robustness with the full set of fixed effects.** Following R6’s request, new Internet Appendix Table IA.19 re-estimates columns 2–10 of Table 7 with industry-year fixed effects, so that every characteristic-control specification is also shown with the full fixed-effect structure. For $1933 \times \tilde{d}$ the estimates are essentially unchanged and remain significant (at the 5% level in seven of nine specifications, 10% in all nine); for $1934 \times \tilde{d}$ significance is lost in seven of

nine columns, mainly through attenuation of the point estimates rather than wider standard errors, with no sign reversals. Section 5 now states this by year and notes that the attenuation appears only when characteristic-period controls are stacked on top of industry-year cells.

- **Limitations of the exposure measure.** Section 5.5 now contains an explicit limitations paragraph acknowledging that, because virtually all corporate bonds of the era carried gold clauses while preferred shares carried none, our exposure measure is necessarily entangled with the composition of a firm’s fixed claims, and that our placebo and control strategies mitigate but cannot fully eliminate this concern.
- **Pre-trends and event-time patterns.** Section 5.2 now acknowledges directly that, given the width of the confidence intervals around individual year estimates, the litigation-year coefficients cannot be statistically distinguished from individual pre-treatment estimates such as the 1930 coefficient, and clarifies that the support for the parallel-trends assumption rests on the full pre-treatment pattern.
- **Liberty bond discussion.** We have revised the historical background section to clarify why Liberty Bond prices rose during the Supreme Court arguments in January 1935. Since Liberty Bonds were gold-denominated government obligations, a government loss would have obligated the Treasury to honor gold payments at the then-prevailing price of \$35 per ounce—69% above the pre-abrogation level—increasing the real value of the bonds to holders. We have also added the missing *New York Times* citation.

Corrections identified while rebuilding the replication code. As part of this revision we rebuilt the paper’s replication code end to end, so that every table and figure regenerates from the raw data. This exercise surfaced two issues in Table 4 of the previous draft, which we have corrected; neither affects any result the paper relies on, and we report both columns before and after the corrections below for completeness. First, the note to column 5 misdescribed the specification: the column uses an alternative exposure measure that scales gold-clause debt by the firm’s total fixed claims (bank debt, bonds, and preferred shares), rather than the baseline $\tilde{d}$ on

a restricted sample. The column header, the table note, and the corresponding text in Section 5 now describe the specification correctly; the estimates are unchanged except for one standard error in the third decimal. Second, the “no redemption” sample of column 4 (and of Internet Appendix Table IA.12, column 1) inadvertently excluded 37 unexposed firms that lacked a prior-year observation during 1931–1935, an artifact of missing-value handling in the original code; the revised sample excludes only the 90 firms that retired their gold-clause debt during the litigation window. The only significance change is that the marginally significant $1928 \times \tilde{d}$ interaction in column 4 loses its 10% star.

	Column 4 (No redemption)		Column 5 (Fixed claims)	
	Round 1	Revised	Round 1	Revised
Q	0.085*** (0.015)	0.086*** (0.014)	0.123*** (0.022)	0.123*** (0.022)
$\tilde{d}$	-0.097** (0.033)	-0.094** (0.034)	-0.057* (0.030)	-0.057* (0.030)
$1926 \times \tilde{d}$	0.057** (0.021)	0.061** (0.022)	0.033 (0.026)	0.033 (0.026)
$1927 \times \tilde{d}$	0.026 (0.028)	0.010 (0.030)	0.043 (0.031)	0.043 (0.031)
$1928 \times \tilde{d}$	0.061* (0.030)	0.039 (0.031)	0.020 (0.028)	0.020 (0.028)
$1929 \times \tilde{d}$	0.046 (0.029)	0.040 (0.029)	0.005 (0.025)	0.005 (0.025)
$1930 \times \tilde{d}$	-0.018 (0.022)	-0.024 (0.022)	-0.038 (0.024)	-0.038 (0.024)
$1931 \times \tilde{d}$	0.004 (0.014)	0.004 (0.014)	-0.015 (0.014)	-0.015 (0.014)
$1933 \times \tilde{d}$	-0.068*** (0.013)	-0.064*** (0.013)	-0.058*** (0.013)	-0.058*** (0.014)
$1934 \times \tilde{d}$	-0.032*** (0.011)	-0.036*** (0.010)	-0.043*** (0.009)	-0.043*** (0.009)
$1935 \times \tilde{d}$	0.025** (0.010)	0.027** (0.010)	0.018** (0.008)	0.018** (0.008)
$1936 \times \tilde{d}$	0.001 (0.012)	0.006 (0.011)	-0.011 (0.012)	-0.011 (0.012)
$1937 \times \tilde{d}$	0.015 (0.016)	0.005 (0.016)	-0.007 (0.017)	-0.007 (0.017)
$1938 \times \tilde{d}$	0.000 (0.019)	0.009 (0.019)	-0.006 (0.019)	-0.006 (0.019)
$1939 \times \tilde{d}$	-0.021 (0.015)	-0.025 (0.016)	-0.034* (0.017)	-0.034* (0.017)
$1940 \times \tilde{d}$	-0.005 (0.016)	-0.013 (0.016)	-0.013 (0.016)	-0.013 (0.016)
R^2	0.221	0.224	0.252	0.252
Observations	5,572	5,918	6,048	6,048

The rebuild surfaced three smaller corrections in the same category. First, the standard errors in column 7 of Table 4 (the bank-debt placebo) were inadvertently clustered by firm only in the previous draft; they are now two-way clustered by firm and year, like every other column. The coefficients are unchanged; two pre-period placebo interactions become significant at conventional levels, and the litigation-year cells remain insignificant. Second, the stock-return series in Section 3 are now built directly from CRSP daily returns; the previous draft's figures came from an inter-

mediate spreadsheet extract that contained an error (the Joint Resolution returns, for example, change from 11.7% and 27.2% to 12.1% and 31.4%), and the post-argument window is now defined by the documented press dates (January 11–17, 1935). Table 1 and the event figures reflect the corrected series. Third, the firm count and summary statistics in Panel C of Table 2 have been corrected (503 firms rather than 594). None of these corrections affects any result the paper relies on.

First, I appreciate the improvement in measuring d_{it} relative to all long-term obligations involving fixed contractual payments, but this only partly addresses my concerns. Preferred shares are not interest bearing, and thus may behave quite differently than debt during downturns or when firms are close to distress. I understand that there is nothing the author can do about this because bank debt is simply too small, but the paper should acknowledge this limitation more clearly and briefly discuss the potential confounders that cannot be addressed due to how firms structured their debt at that time.

The referee raises a valid point, and we now address it explicitly in the manuscript. Preferred dividends are discretionary in a way that bond coupons are not—deferring them does not trigger default—and we agree that this distinction matters in distress, even though the flexibility was partial, as preferred issues of the era were typically cumulative. Because preferred shares enter the denominator of $\tilde{d}$ but not the numerator, the concern is that high- $\tilde{d}$ firms, which rely more on inflexible bond financing, may be more vulnerable to downturns generally, so that the 1933–1934 differential partly reflects debt rigidity rather than gold clause litigation.

Two features of the evidence speak to this concern, though we acknowledge at the outset that neither can resolve it entirely. The preferred-share placebo (column 6 of Table 4) replaces the numerator of $\tilde{d}$ with preferred shares and finds positive, insignificant 1933 and 1934 coefficients, whereas a general flexibility effect would tend to push these coefficients negative. And the temporal pattern (Figure 3) is concentrated in 1933–1934 and reverses after the 1935 ruling, which is difficult to reconcile with debt rigidity alone: that mechanism would also operate in the 1937–1938 recession, where we find no differential effect. The industry-year fixed effects and characteristic-interacted controls of Table 7 provide further protection against shocks correlated with industry and observable firm structure.

These tests mitigate but cannot fully eliminate the concern. Section 5.5 now closes with an explicit limitations paragraph stating this; it reads in part:

“[B]ecause virtually all corporate bonds of the era carried gold clauses while preferred shares carried none, $\tilde{d}$ is entangled with the composition of a firm’s fixed claims: pre-

ferred shares enter its denominator but not its numerator, so preferred-heavy firms have mechanically lower measured exposure. Because preferred dividends are discretionary and non-interest-bearing, such firms may also behave differently in distress, raising the concern that $\tilde{d}$ partly captures a general sensitivity to downturns among bond-reliant firms . . . An ideal design would vary gold-clause exposure while holding the debt-versus-preferred composition fixed; the institutional structure of the period precludes this.”

I appreciate using 1936–37 as a placebo. While this helps address concerns about downturns driving the results, there are other important macro differences between both periods—such as the collapse of bond issuance, that limited rollover—that may affect exposed firms more in 1933–34 beyond the uncertainty of the abrogation of the gold clause. That is, the placebo helps but only for a subset of the concerns about confounders. To that end, the robustness regressions are more helpful. Though the findings are broadly reassuring, they do point to either the thinness of the data or a lack of robustness. The authors cannot control for industry $\times$ year and $X_{it} \times$ year at the same time, so the fixed effects vary across specifications in Table 6. The paper argues this is done “To assess robustness to firm-level controls separately from industry-level controls, this specification omits industry-year fixed effects.” This sounds like a copout. Given the concerns of New Deal policies as confounders, which may have affected firms differently given their characteristics, one may want to control for both. My sense is that they lack power, as even without the industry-year FEs the significance is lost for $1934 \times d$ across several specifications for Columns 2–10. I would like to see the results with industry-year FEs for these columns as well. If significance is lost across the board, then the paper should acknowledge it and make its best case to defend their preferred specifications due to limited power—but readers should know whether the effects are sensitive so they can pass their own judgement.

We thank the referee for these comments and address each in turn.

Placebo and the rollover channel. The referee correctly notes that the 1936–37 placebo addresses recession-driven confounders but not the collapse of bond issuance that may have disproportionately

affected gold-exposed firms—which held more bonds—through a rollover channel (the collapse in industrial bond issuance is shown in Figure 4 of the paper). Column 3 of Table 4 speaks to this concern without fully resolving it: it excludes firms with bonds maturing at any point between 1931 and 1934, and the 1933 and 1934 coefficients remain -0.058 and -0.037 , essentially unchanged from the baseline, suggesting that the investment effect is not entirely driven by firms with immediate refinancing needs. We also say this in the paper now: Section 5.2 notes that “[t]his placebo ... speaks only to recession-driven confounders. Conditions unique to the litigation years, such as the collapse of bond issuance and the rollover risk it created, have no 1937–1938 counterpart; the sample-restriction tests below address them directly.”

Industry-year fixed effects in columns 2–10. We agree that readers should be able to assess this sensitivity directly. Internet Appendix Table IA.19 re-estimates columns 2–10 of Table 7 (Table 6 in the previous draft) with industry-year fixed effects in place of year fixed effects, so that all ten specifications share the fixed-effect structure of column 1; its columns (1)–(9) correspond to columns 2–10 of Table 7.

The referee asked whether the effects become noisy or whether they become small and insignificant. The answer differs by year, and Section 5 now states it in exactly these terms. For $1933 \times \tilde{d}$, the estimates are essentially unchanged and remain significant—at the 5% level in seven of nine specifications and at the 10% level in all nine. For $1934 \times \tilde{d}$, significance is lost in seven of nine specifications, and the loss comes mainly from attenuation of the point estimates (which range from -0.012 to -0.031 , against -0.022 to -0.043 under year fixed effects) rather than from wider standard errors. The estimates never reverse sign, and the attenuation appears only when characteristic-period controls are stacked on top of industry-year cells: column 1 of Table 7, which absorbs industry-year fixed effects without characteristic controls, estimates $1934 \times \tilde{d}$ at -0.036 , significant at the 5% level.

The attenuation has a mechanical source. Industry and firm characteristics overlap—e.g., steel firms are large and levered—so the characteristic-period dummies and the industry-year cells compete for the same variation; the 1934 estimate attenuates and crosses conventional thresholds first. In

the all-controls column the overlap is exact—several characteristic-period interactions are dropped outright for multicollinearity, as Section 5.5 notes—which is why the 1933 interaction there (and in the cash-decile column) retains significance only at the 10% level.

Column 1 of Table 7 remains our preferred specification for the New Deal confounder concern: industry-year fixed effects absorb any shock common to an industry in a given year. The characteristic controls address the distinct concern that gold exposure proxies for observables on differential investment trajectories, and they continue to do so under industry-year fixed effects. Combining the two exercises attenuates the 1934 estimate; we say so explicitly in Section 5 and provide Table IA.19 so that readers can judge the sensitivity themselves.

The authors made no comment about the fact that the 1933 and 1934 coefficients do not seem statistically different than the 1930 effects (Figure 3).

The referee is right: the 1933 and 1934 coefficients cannot be statistically distinguished from the 1930 coefficient. We now acknowledge this directly in the paper; Section 5.2 adds:

“We acknowledge that, given the width of the confidence intervals around individual year estimates, the 1933 and 1934 coefficients cannot be statistically distinguished from individual pre-treatment estimates such as the 1930 coefficient; the support for the parallel-trends assumption rests on the full pre-treatment pattern.”

That pattern—pre-treatment coefficients that are uniformly small and individually insignificant, with 1931, the last estimated pre-treatment year, essentially at zero relative to the omitted 1932 base—is what the design relies on.

One small comment—my understanding (though I could be wrong) is that repurchases in the open market were not allowed post 1929. I imagine the main reason driving the decline in shares was reorganizations. If that is the case, it is not obvious one would want to include this if the point of the analysis is to estimate whether firms were using payout policy to preemptively give the cash to shareholders. That said, the results in Table 4 show that the effects are driven by dividends, not

repurchases, so this is mostly about guiding modern readers on what firm policies were in the 1930s.

The referee’s comment prompted us to examine this directly in our data. Two facts stand out. First, treasury stock does appear on the Moody’s balance sheets in our sample: between a fifth and a third of firms report a nonzero treasury-stock position during 1930–1934, and net additions to treasury stock occur in roughly one in eight firm-years. We cannot tell from the balance sheet how these shares were acquired—open-market purchases, private transactions, exchanges, or settlements—so this observation does not by itself settle the institutional question the referee raises. Whatever their source, the amounts are small: the median net addition is about 0.5% of total assets. Second, our net repurchase measure is built from split-adjusted changes in shares outstanding, and over 1930–1934 those changes rarely reflect share reductions. The referee’s economic point therefore stands regardless of the institutional details: nothing in this period resembles the discretionary buyback programs of modern data, and net repurchases were not a meaningful payout vehicle.

We agree that modern readers would benefit from this context, and we have added a footnote to Section 5 stating these facts: treasury-stock additions occur but are small, the number of shares outstanding is typically unchanged, and the variation in the net repurchase measure comes mostly from share issuance rather than retirement.

This does not affect any of our substantive conclusions. As the referee notes, Table 5 (Table 4 in the previous draft) shows that the payout effect is driven entirely by cash dividends (column 2 is significant), while the net repurchase coefficient (column 3) is statistically insignificant in 1933 and 1934. The channel relevant to our debt overhang interpretation—shareholders extracting value through discretionary distributions in anticipation of losing their claim on the assets—operates through dividends, not through changes in shares outstanding.

Additional comment (cover letter)

A key issue is whether the authors can clearly disentangle the effects of uncertainty about the abrogation of gold clauses from other confounders—other policies and firm characteristics, and whether it is about the gold clause vs interest-bearing leverage. There is not much they can do about

the second given how firms structured their debt. I appreciate the refinement in the definition of the treatment effect, but the ideal case would be not to have preferred shares either. I am willing to look passed this, but the paper should be more upfront. The bigger issue is other confounders. They have tried to address this in the robustness section in Table 6, but as you can see they cannot control for a full set of fixed effects. What bothers me more is that they have not shown us the results with all effects, and instead state they do not include industry-year FEs in all columns to assess the robustness to industry and firm separately. This is a weak argument. If the issue is that they have too few firms to do this without the treatment effects being killed, they should say that. Readers may understand, but they need to know. And I would like to see how bad effects get—is it noisy or do the results become small and insignificant? I also find the placebo less helpful than the paper claims, but that’s a personal taste.

We are grateful for the referee’s overall assessment and for the clarity of this summary. We agree with the central message—that the paper should be upfront about what the data can and cannot establish—and we have revised the manuscript accordingly. We respond to the two main issues in turn, and indicate where each is now addressed in the paper.

Gold clause exposure versus interest-bearing leverage (preferred shares). The referee is correct that the ideal design would isolate gold-clause exposure while holding the debt-versus-preferred composition fixed, and that the institutional structure of the period does not permit this: virtually all corporate bonds of the era carried gold clauses, while preferred shares carried none. We no longer leave this implicit. Section 5.5 now contains an explicit limitations paragraph stating that $\tilde{d}$ is necessarily entangled with the composition of a firm’s fixed claims, that preferred-heavy firms may behave differently in distress for reasons unrelated to gold clauses, and that our tests mitigate but cannot fully eliminate this concern.

Other confounders and the full set of fixed effects. We agree that the previous draft’s justification for not always including industry-year fixed effects read as a rationalization, and we have removed that framing. We now do two things. First, we state the real reason plainly in Section 5.5: with a sample of this size, absorbing industry-year variation and characteristic-period variation simultaneously

exhausts the identifying variation available to estimate the treatment effects. Second, and most importantly, we now show the results. Internet Appendix Table IA.19 re-estimates all of columns 2–10 of Table 7 with industry-year fixed effects, so the reader sees exactly “how bad effects get.” The answer differs by year. For $1933 \times \tilde{d}$, the estimates are essentially unchanged and remain significant—at the 5% level in seven of nine specifications and at the 10% level in all nine. For $1934 \times \tilde{d}$, significance is lost in seven of nine specifications, and the loss comes mainly from attenuation of the point estimates (which range from -0.012 to -0.031 , against -0.022 to -0.043 under year fixed effects) rather than from wider standard errors. The estimates never reverse sign, and the attenuation appears only when characteristic-period controls are stacked on top of industry-year cells: column 1 of Table 7, which absorbs industry-year fixed effects without characteristic controls, estimates $1934 \times \tilde{d}$ at -0.036 , significant at the 5% level. The effect does not reverse or vanish, and we now say so directly rather than asking the reader to take it on faith.

On the placebo. We take the referee’s point that the 1936–37 placebo addresses only a subset of confounders—chiefly recession-driven ones—and not, for example, the collapse of bond issuance (Figure 4 of the paper) and the attendant rollover risk. We have tempered the paper’s language about the placebo accordingly and now lean more heavily on the direct robustness checks, including the exclusion of firms with bonds maturing during the litigation window (column 3 of Table 4), which speaks specifically to the rollover channel.

Taken together, these revisions are intended to make the paper’s limitations visible to readers while preserving what we believe is the core, robust finding: gold-exposed firms reduced investment specifically during the 1933–1934 litigation window and reversed afterward, a pattern that survives industry-year fixed effects in our preferred specification and that is not reproduced by the preferred-share or bank-debt placebos.

REFERENCES